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Smart phishing detection system: Using Machine Learning Model Integrated with Virustotal API.

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Abstract:
Phishing is one of the most important and challenging areas in the spectrum of cybersecurity, always adapting to overcome traditional techniques of detection. We present the Smart Phishing Detection System based on reinforcement learning coupled with the VirusTotal API to improve the identification of harmful links and executables. In essence, our platform allows users to upload any suspicious link or file recognized as phishing, and then analyzes it via a machine able to apply VirusTotal API functionality. The reinforcement learning mechanism is responsive to new techniques being deployed in the phishing space; hence, with time, improving the precision of detection. Furthermore, it provides users with a comprehensive view of how different antivirus vendors perceive the file or URL as malicious to ensure transparency in threat assessment. The approach tackles the weaknesses of traditional signature detection by adopting a flexible, data-centric approach. The system has been verified with sufficient testing against a large variety of phishing threats, thus yielding real-time responses with great accuracy. In particular, the task is designed to assist both individual end-users as well as cybersecurity professionals in identifying and responding to phishing threats efficiently, thereby increasing the overall awareness of various cyber threats.

Keywords: Phishing detection, Reinforcement learning, VirusTotal API, Machine learning, Cybersecurity, Threat analysis, Hybrid detection system, URL classification, Antivirus vendors, Real-time detection.

I. Introduction:

It is so clear that phishing is one of the greatest threats to both individual and organizational operations all over the globe. New phishing strategies are now developed and become more sophisticated that could easily bypass detection by traditional phishing detection systems. In addition, attacks always take the form of misleading emails, fraudulent websites, or malicious executable files with the aim of stealing some sensitive data, deploying malicious software, or causing harm to users. Recent findings show that there is an increasing frequency and sophistication of phishing attacks that create the necessity of having improved detection schemes.

Traditional phishing detection systems primarily depend on signature-based techniques which certainly fail to recognize most of the emerging threats. These challenges could be turned to reality and thus trends of machine learning (Ml) methods were introduced for the real time phishing detection used for more adaptive data-driven solutions. Amongst all methods of ML, reinforcement learning has been effective since the models will always be adapting to the changing data and patterns. This flexibility is important especially in the field of cybersecurity since the modalities utilized by the attackers keep changing.

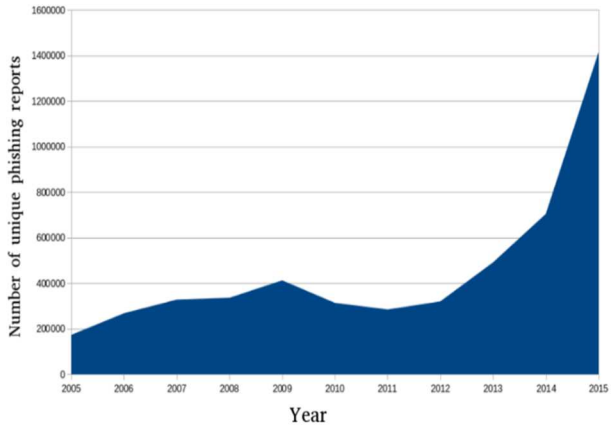

Fig. 1. Phishing reports from 2005 - 2015

This paper introduces a Smart Phishing Detection System that integrates reinforcement learning with the VirusTotal API to detect malicious URLs and executable files. VirusTotal is a reputable platform that compiles results from numerous antivirus engines, offering valuable insights for threat

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identification. By merging the VirusTotal API with a reinforcement learning framework, our system provides strong, real-time detection capabilities while also visualizing insights from different vendors. Users can submit URLs or EXE files to the platform and receive prompt feedback regarding potential threats, along with a detailed analysis of which vendors have flagged the content as malicious.

II. Literature survey

One of the common methods hackers are using today to test weaknesses in internet security is through phishing attacks. Factual proof indicates that many detection methodologies have been put forward based on the use of ML models coupled with APIs to find out these vulnerabilities. For example, in [1], a machine learning model was developed to classify phishing websites on 22 ranks of URL features, producing very high accuracy. The core features that played a considerable role in the performance of the model included general specific characters, SSL certificates, and Google indexing. Advanced algorithms such as XGBoost were implemented in [2] to further identify phishing sites.

In another research endeavor [3], the authors aimed to put to test phishing predictions via incorporation with the VirusTotal API. This one scans for malicious behavior in the submitted URLs and files, compares results with those of many security vendors and provides relatively precise results. Hybrid techniques were also tested in [4] such that machine learning model derived predictions merged with VirusTotal data produced further benefits. The two tiered checking method thus led to better accuracy in detection.

Therefore, further studies mainly concentrated on the integration of frontend and users. For example, [5] addressed a user interface based on React that allowed users to submit URLs and files for analysis. This gave live visualization such as pie charts showing how many vendors tagged a link as malicious. And the feature on download with PDFs having results and visualization analyses in created better information accessibility to end users [6].

Phishing met simulators and User-friendly interfaces that matter were emphasized in research [7]. It also had a web application similar to such VirusTotal where users could submit links for phishing assessment and use better visualization techniques such as interactive charts and extensive assessments to report in detail. A typical example is that in [8], where users were able to upload their files to check through the VirusTotal API, having a comprehensive overview of phishing detection.

As mentioned in [9] and [10], many research works have hitherto sought other alternatives to the conventional backends such as Flask, where different frameworks were evaluated for better scalability and better handling of errors. To improve the interfacing between front-end and back-end, [11] offered a long procedure for making API integrations to ensure the best deployment of phishing detection systems.

Real-time phishing detection approaches applied on social networking applications were included in [12]. URL analysis leveraged the power of LSTM and ANN models to achieve these applications.

III. Proposed techniques:

To identify phishing URLs and malicious files in a systematic manner, a hybrid model has been proposed. System architecture along with its functions has been discussed in detail in Fig. O. The platform includes medians to give machine learning and virustotal files to ensure that appropriate and detailed mechanisms are followed in threat detection. Twenty-three features have been introduced from URLs to detect phishing. Further, it involves an amalgamation of many operational phases: URL/File Input, Feature Extraction, Machine Learning Prediction, VirusTotal Integration, Threat Assessment and Reporting. It also involves external vendor-based analyses integrated into predictive models, greatly increasing accuracy as well as relevance in identifying aspiring cybersecurity threats.

A. The collection of the data.

The data from this research was gathered from Kaggle. There was a total of about 87 features in the entire dataset. However, only 23 from the original 87 were selected in consideration of this work. The subset study based on the 23 features is attached. The dataset comprises 11,430 records such that there is a random 50% representation for legitimate websites and another 50% for URLs. The training for these four machine learning models would be run on this dataset.

Table.1. Dataset Explanation

Feature	**Explanation**	**Type**
Length_url	Total length of tokens in the URL.	Numeric
Length_Hostname	Length of the hostname of URL	Numeric
IP	Indicates if hostname has IP in URL	Binary
No_of_dots	Number of dots in URL	Numeric
No_of_hyphens	Number of hyphens in URL	Numeric
No_of_at	Number of '@' in the URL	Numeric
No_of_qn	Number of '?' in the URL	Numeric
No_of_And	Number of '&' in the URL	Numeric
No_of_Or	Number of '|' in the URL	Numeric
No_of_Eq	Number of '=' in the URL	Numeric
No_of_Underscore	Number of '_' in the URL	Numeric
No_of_Tilde	Number of '~' in the URL	Numeric
No_of_Percent	Number of '%' in the URL	Numeric
No_of_Slash	Number of '/' in the URL	Numeric
No_of_Star	Number of '*' in the URL	Numeric
No_of_Colon	Number of ':' in the URL	Numeric
No_of_Comma	Number of ',' in the URL	Numeric

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No_of_Semicolon	Number of ';' in the URL	Numeric
No_of_Dollar	Number of '$' in the URL	Numeric
No_of_Space	Number of ' ' in the URL	Numeric
No_of_www	Number of 'www' in the URL	Numeric
No_of_Com	Number of 'com' in the URL	Numeric
No_of_Dslash	Number of '\' in the URL	Numeric
http_in _path	Indicates the presence of "http" within the URL path.	Binary
https_token	Indicates the presence of "https" within the domain or path.	Binary
Dig_url	Ratio of numeric characters to the overall length of the URL.	Numeric
Dig_host	The ratio of numerical digits to the total length of the hostname.	Numeric

B. Machine learning models

The notorious phishing attacks targeting websites were discernibly detected using four classifiers-XGBoost, Support Vector Machine(SVM), Random Forest and K-Nearest Neighbors and trained with appropriate features from URLs to evaluate each with standard classification matrices such as accuracy, F1-score and precision . Consequentially, every model was investigated for its capacity to differentiate between phishing and legitimate URLs, and the findings represented in a grouped bar chart for comparative analysis.

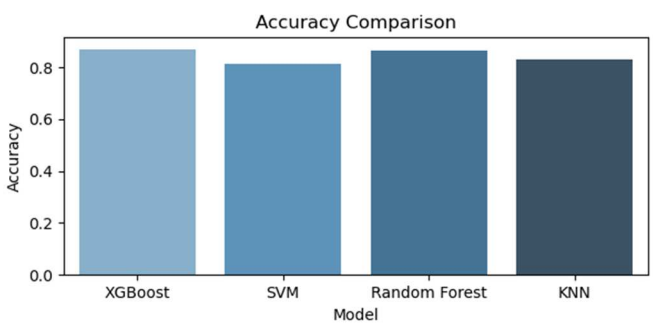

Fig . 2. Accuracy Comparison

a) K-nearest neighbors – KNN

KNN or k-nearest neighbors is a non-parametric, very simple learning instance-based algorithm which assigns data points based on their closeness to their k-nearest neighbor points in training dataset. New data point gets classified using the class that obtains the maximum among its neighbors. It mainly works for a smaller dataset, but whenever the size of data increases, then it gets very expensive.

Table. II. KNN report

	P	R	F1	S
0	0.82	0.84	0.83	1157
1	0.83	0.82	0.83	1129
accuracy	0.83	0.83	0.83	0.83
macro avg	0.83	0.83	0.83	2286
weighted avg	0.83	0.83	0.83	2286

b) Support vector machine -SVM

Support Vector Machine (SVM) is a supervised learning technique that identifies the most suitable hyperplane for partitioning data into separate categories. By employing various kernel functions, such as linear, polynomial, or radial basis function (RBF), SVM is capable of addressing both linearly and non-linearly separable datasets. This algorithm demonstrates significant efficacy in high-dimensional environments and emphasizes the maximization of the margin between different classes, thereby enhancing the accuracy of classification outcomes.

Table. III. SVM report

	P	R	F1	S
0	0.83	0.79	0.81	1157
1	0.80	0.84	0.82	1129
accuracy	0.81	0.81	0.81	0.81
macro avg	0.82	0.81	0.81	2286
weighted avg	0.82	0.81	0.81	2286

c) Random forest -RF

Random Forest is an ensemble learning algorithm that builds multiple decision trees and aggregates their predictions to enhance accuracy and reduce overfitting. It operates by training each tree on a random subset of features and data samples. The final prediction is made through a majority vote (classification) or averaging (regression), making Random Forest robust and interpretable.

Table. IV. RF report

	P	R	F1	S
0	0.85	0.89	0.87	1157
1	0.88	0.84	0.86	1129
accuracy	0.87	0.87	0.87	0.81
macro avg	0.87	0.87	0.87	2286
weighted avg	0.87	0.87	0.87	2286

d) Extreme Gradient Boosting -XG Boost

XGBoost embodies a robust and high-performing implementation of a gradient boosting framework. This technique assembles a number of decision trees, continually improving the loss function and correcting mistakes made in the previous iterations. Indeed, efficiency and speed are hallmarks of this algorithm, which bestows upon XGBoost an increased competency in handling imbalanced and sparse datasets compared to other algorithms and also regularization of models, thus minimizing overfitting risks usually caused by conventional boosting approaches. Such a performance, as well as its joint capability with support vector machines (SVM), has further improved its accuracy rates, especially while being deployed for phishing prediction.

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Table. V. XG boost report

	P	R	F1	S
0	0.86	0.88	0.87	1157
1	0.88	0.86	0.87	1129
accuracy	0.87	0.87	0.87	0.81
macro avg	0.87	0.87	0.87	2286
weighted avg	0.87	0.87	0.87	2286

C. VirusTotal

VirusTotal is a well-respected online utility that gathers results from numerous antivirus and security vendors and assesses a given URL or file against the threat level. It scans by comparing files or URLs with the complete databases of recognized malicious content to deliver real-time assessments from several security vendors. In phishing detection, VirusTotal draws on a huge plethora of threat intelligence sources to analyze URLs to check if there are links to phishing activity, malware distribution, and other nefarious acts. In a similar instance for file checking, VirusTotal performs a multi-vendor scan in search of known malware or unwanted code within files. The service creates thorough reports for any analysis that includes the results from different security engines as experiences show whether a file or URL is clean, malicious, or suspicious. Security systems are improved when they are feed insights from VirusTotal for an enhanced assessment of threats and providing users with a better level of confidence when assigning risk to dangerous URLs or files.

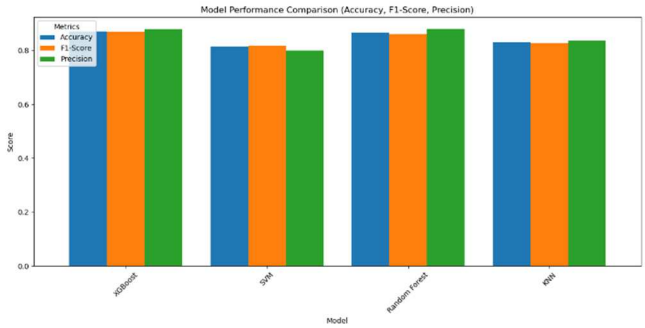

Fig. 3. Accuracy Comparison

D. URL Detection

URL analysis with an integrated machine learning model and VirusTotal is used to identify potential phishing threats. When a user submits a URL, it is passed through a feature extraction function, which retrieves 23 types of characteristics specific to the URL, such as the length of URL, presence of suspicious characters, as well as structural patterns of the domain configuration and keywords that have significant affinity with phishing sites. After extraction, feature compilation occurs in a feature matrix in an ordered form. This matrix together with a target scalar, which declared on whether the URL is legitimate or phishing, is used to train multiple machine learning models including XGBoost which would be able to predict whether this URL would be safe based on identified features. This submission is simultaneous to VirusTotal, widely recognized for running URLs against an incredibly broad-based database of known malicious URLs. The context beyond current threats, however, could also be found in the response the user received such as 'malicious', 'clean', or 'underrated' from several security vendors. Thus, the user will see the results of both mechanisms on the webpage, providing a complete assessment. When these URLs flag as valid, the user can proceed with their activity of web browsing, and the false ones would be blocked for a secure online experience.

E. File Handling

The file analysis part of this project allows the users to upload their files for inspections against security threats. On the uploading of a file, it is immediately sent to VirusTotal, which is based on a whole lot of databases and has a record of known malicious files. VirusTotal compares the uploaded files against results from many antivirus engines and security providers. This is so that the system detects if it contains any viruses, malware, trojans, or any other harmful content that may endanger the end user's system. It has been scanned by varied antivirus vendors, all of whom have their unique insights towards the file's safety. Post this scanning; VirusTotal prepares a full report stating whether the file is clean, malicious, or suspicious by the various engines of the security scans. The results are then routed to your system, which presents them on the webpage for end-users to view what different specific security vendors say. The user is informed that the file is safe for use if it is considered clean. If the file is otherwise malicious or suspicious, an immediate warning is given to the user pertaining to the possible risk, whereas it's flagged as unsafe. Such integration with VirusTotal guarantees that every single uploaded file goes through tremendous scrutiny by several reputable security vendors, giving therefore a reliable and comprehensive security report before the end-user interacts with the file.
Your project, thus, incorporates VirusTotal's multi-faceted scanning outcomes making a huge difference in detecting harmful files consequently ensuring the thorough informing of users on possible threats concerning the files uploaded.

IV. Conclusion

Phishing Attack is one of the ongoing problems of Cyber Security, with personal and financial data worth stealing. The latest research is focused on an innovative and modern phishing detection system that integrates machine learning techniques and outside threat intelligence so that it aids in filling the security gaps. In this regard, the study considered some popular algorithms, like XGBoost, K-Nearest Neighbors (KNN), Random Forest, and Support Vector Machines (SVM). From all these models, testing showed that XGBoost

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was the best model with an overall accuracy of 87%, given a balanced precision, recall, and F1-scores, while identifying phishing URLs. It is thus the most appropriate model for this type of application, owing to its predictive capability.

This approach uses a unique feature extraction process that focuses highly on the structural, behavioral, and statistical aspect of the URLs that could reveal a phishing attempt. These also include scans from VirusTotal's multi-vendor analyses in real-time threat verification, making it even more robust. Coupling the existing machine learning algorithm and external threat intelligence provides a two-layer security measure, thus enhancing detection of phishing attacks.

Because of the serious implications of phishing attacks such as loss of data or financial loss, these hybrid strategies emphasize the need of such systems for conducting advanced detection methodologies in any cyber-defense architecture. Such results stand nice verifications regarding the strength of machine learning combined with threat intelligence in devising a solid anti-phishing detection system.

The present research, however, enhances the functioning of the developed system by including some user-related features like a facility for reporting files as well as suspicious URLs. This would assure that users should also be empowered to play an active role against phishing threats whereby the overall impact of such attacks is then reduced and thus strengthens cybersecurity.

Reference

[1] Gokul Rangasamy, Primya T, Harinath Sivakumar, and Kanagaraj G. *"PHISHION: Phishing Detection Application."* Proceedings of the 2024 Second International Conference on Emerging Trends in Information Technology and Engineering (ICETITE), Coimbatore, Tamil Nadu, India. IEEE, 2024.

[2] A.A. Orunsolu, A.S. Sodiya, and A.T. Akinwale. *"A Predictive Model for Phishing Detection."* Journal of King Saud University - Computer and Information Sciences, vol. 34, pp. 232–247, 2022.

[3] Emmanuel Song Shombot, Gilles Dusserre, Robert Bestak, and Nasir Baba Ahmed. *"An Application for Predicting Phishing Attacks: A Case of Implementing a Support Vector Machine Learning Model."* Cyber Security and Applications, vol. 2, Article 100036, 2024.

[4] Sakib Shahriar Shafin. *"An Explainable Feature Selection Framework for Web Phishing Detection with Machine Learning."* Data Science and Management, vol. 116, 2024.

[5] Nguyet Quang Do, Ali Selamat, Hamido Fujita, Ondrej Krejcar. *"An Integrated Model Based on Deep Learning Classifiers and Pre-trained Transformer for Phishing URL Detection."* Future Generation Computer Systems, vol. 161, 2024, pp. 269–285.

[6] Antonio Paya, Sergio Arroni, Vicente García-Díaz, Alberto Gómez. *"Apollon: A Robust Defense System Against Adversarial Machine Learning Attacks in Intrusion Detection Systems."* Computers & Security, vol. 136, 2024, 103546.

[7] Grega Vrbančič, Iztok Fister Jr., Vili Podgorelec. *"Datasets for Phishing Websites Detection."* Data in Brief, vol. 33, 2020, 106438.

[8] R. Jayaraj, A. Pushpalatha, Deepa Damodaran. *"Intrusion Detection Based on Phishing Detection with Machine Learning."* Measurement: Sensors, vol. 31, 2024, 101003.

[9] Nayanamana Samarasinghe, Mohammad Mannan. *"On Cloaking Behaviors of Malicious Websites."* Computers & Security, vol. 101, 2021, 102114.

[10] SK Hasane Ahammad, E Venkatesh Babu, Sunil D. Kale, Gopal D. Upadhye, Amol V. Dhumane, Sandeep Dwarkanath Pande, Dilip Kumar Jang Bahadur. *"Phishing URL Detection Using Machine Learning Methods."* Advances in Engineering Software, vol. 173, 2022, 103288.

[11] Asadullah Safia, Satwinder Singh. *"A Systematic Literature Review on Phishing Website Detection Techniques."* Journal of King Saud University – Computer and Information Sciences, vol. 35, no. 2023, pp. 590–611.

[12] Manuel Sánchez-Paniagua, Eduardo Fidalgo, Enrique Alegre, Rocío Alaiz-Rodríguez. *"Phishing Websites Detection Using a Novel Multipurpose Dataset and Web Technologies Features."* Expert Systems With Applications, vol. 207, 2022, p. 118010.

[13] Sultan Asiri, Yang Xiao, Saleh Alzahrani, Tieshan Li. *"PhishingRTDS: A Real-Time Detection System for Phishing Attacks Using a Deep Learning Model."* Computers & Security, vol. 141, 2024, p. 103843.

[14] Mohammed M. Alani, Hissam Tawfik. *"PhishNot: A Cloud-Based Machine Learning Approach to Phishing URL Detection."* Computer Networks, vol. 218, 2022, p. 109407.

[15] Choon Lin Tan, Kang Leng Chiew, Kok Sheik Wong, San Nah Sze. *"PhishWHO: Phishing Webpage Detection via Identity Keywords Extraction and Target Domain Name Finder."* Decision Support Systems, vol. 88, 2016, pp. 18-27.

[18] Hossein Shirazi, Kyle Haefner, Indrakshi Ray. *"Fresh-Phish: A Framework for Auto-Detection of Phishing Websites."* Department of Computer Science, Colorado State University, Fort Collins, USA.

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